ASSIGNMENT 4

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import fetch_openml
boston = fetch_openml(name="boston",version=1)

data = pd.DataFrame(boston.data)

data.columns=boston.feature_names

data.head()

data['PRICE'] = boston.target

data.isnull().sum()

x = data.drop(['PRICE'], axis = 1)
y = data['PRICE']

from sklearn.model_selection import train_test_split
xtrain, xtest, ytrain, ytest =train_test_split(x, y, test_size =0.2,random_state = 0)

import sklearn
from sklearn.linear_model import LinearRegression
lm = LinearRegression()
model=lm.fit(xtrain, ytrain)

xtrain = xtrain.astype(float)
```

```
xtest = xtest.astype(float)

ytrain_pred = lm.predict(xtrain)
ytest_pred = lm.predict(xtest)

df=pd.DataFrame(ytrain_pred,ytrain)
df=pd.DataFrame(ytest_pred,ytest)

from sklearn.metrics import mean_squared_error, r2_score
mse = mean_squared_error(ytest, ytest_pred)
print(mse)
mse = mean_squared_error(ytrain_pred,ytrain)
print(mse)

mse = mean_squared_error(ytest, ytest_pred)
print(mse)

plt.scatter(ytrain ,ytrain_pred,c='blue',marker='o',label='Training data')
plt.scatter(ytest,ytest_pred ,c='lightgreen',marker='s',label='Test data')
plt.xlabel('True values')
plt.ylabel('Predicted')
plt.title("True value vs Predicted value")
plt.legend(loc= 'upper left')
plt.plot()
plt.show()
```